

Bulk-color mechanism mapping v0.1 — what’s needed, what’s constrained, what’s hard. Lays out the joint #252/#414/quark-flip target without claiming a derivation. Holds Keeper’s calibration: two structurally distinct  $B_2$ -specific 3-folds (color from  $h^\vee$ ; spinor<sup>3</sup>-channels from bosonic fundamentals count) that COINCIDE NUMERICALLY — the unification CLAIM requires deriving spinor<sup>3</sup> from  $h^\vee$ .

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## Bulk-color mechanism mapping v0.1

### 0. Why this is the joint target

Three open items converge on one mechanism: - #252 (route II): the colorless-projection mechanism for generations — needs the substrate’s color structure to project from. - #414 v0.2 burden: how does ONE  $h^\vee=3$  produce TWO INDEPENDENT 3-fold structures (color  $\times$  generations = 9 quark combinations)? - #416 v0.2 (quark/gauge per-particle flip): blocked on the substrate’s color mechanism, since  $SU(3) \not\cong K$ .

All three need the same mechanism: how does the substrate produce a 3-fold “color” structure that’s NOT in  $K$ , and how is that structure independent of generations? This doc maps that mechanism’s problem space — without claiming a derivation.

### 1. Constraints the mechanism must satisfy

| label | constraint                                                                                                                                                 |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)   | 3-fold (observed: 3 quark colors)                                                                                                                          |
| (b)   | lives in the BULK of $D_{IV}^5$ (NOT in $K=SO(5)\times SO(2)$ )<br>— RIGOROUS: $SU(3)=A_2 \neq B_2=SO(5)$ , rank-2 algebras differ, so $SU(3) \not\cong K$ |
| (c)   | CROSS-CUTTING with generations (3 colors $\times$ 3 generations = 9 quark combinations physically real)                                                    |
| (d)   | emergent $SU(3)$ gauge phenomenology (low-energy QCD; confinement; asymptotic freedom)                                                                     |
| (e)   | consistent with $N_c = h^\vee(B_2) = 3$ as the substrate origin of the count                                                                               |

The (b) constraint is the structural anchor (a Lie-algebra fact, not a hypothesis). The (c)+(d) constraints are the phenomenological tests. The (e) is the link to the substrate’s derived quantity.

## 2. Available bulk structure (what the dictionary doesn't use)

$D_{IV}^5 = SO_0(5,2)/[SO(5)\times SO(2)]$ . Beyond K, the bulk carries:

- Non-compact piece  $\mathfrak{p}$ :  $\mathfrak{so}(5,2) = \mathfrak{k} \oplus \mathfrak{p}$ ,  $\dim \mathfrak{p} = 21 - 11 = 10$ . As a K-rep,  $\mathfrak{p} = 5_{+1} \oplus 5_{-1}$  (the holomorphic + antiholomorphic tangent space at the origin). This is the bulk's geometric content K doesn't see directly.
- Bergman kernel: singularity exponent  $= \rho_1 = 5/2 = \text{Hua genus} = \text{lepton Casimir (L4 alignment)}$ . Governs Hilbert-space structure on holomorphic functions.
- Holomorphic discrete series of  $SO(5,2)$  on the bulk: K-type decomposition with multiplicities, lowest K-type = trivial (Higgs vacuum), tower structure from the radial direction.
- Wallach / Berezin discrete set: rank-controlled discrete points where the discrete series sits.
- $\text{rank}(SO(5,2)) = 3$  (vs  $\text{rank}(K) = 2+1 = 3$ ) — the non-compact directions add no new Cartan rank, but they add 10 generators.

So the bulk-structural data IS rich (10-dim non-compact piece, Bergman kernel, holomorphic discrete series), but none of it presents an obvious 3-fold or  $SU(3)$ .

## 3. Three candidate handles for the 3-fold (each with a “why it might work” and a “why it's hard”)

### 3.1 $h^\vee = 3$ as substrate counting (Track P-adjacent)

Idea:  $N_c = h^\vee(B_2) = 3$  is a derived substrate primary (Grace + earlier work). “Color” is the manifestation of this 3-fold counting in some bulk operator/sector.

Why it might work: directly satisfies (e); a counting-not-symmetry mechanism is consistent with BST's pattern of deriving structures from substrate primaries rather than positing gauge groups.

Why it's hard: needs to be specified WHERE the counting acts (which operator/sector carries  $h^\vee$  in the bulk) and HOW it produces emergent  $SU(3)$  gauge phenomenology (d). Abstract counting  $\rightarrow$  concrete bulk structure is the missing step.

### 3.2 Spinor<sup>3</sup>-channels (Elie E7)

Idea:  $\text{spinor} \otimes \text{spinor} \otimes \text{spinor}$  decomposes with multiplicity 3 for the spinor (= 3 E6 channels: trivial / vector / adjoint intermediate). Elie proposed this as the generation mechanism. By analogy or symmetry, a similar self-fusion structure could carry color.

Why it might work:  $B_2$ -specific ( $A_2$  gives 0), rigorous Racah-Speiser, dim-validated. Connects to the fermion-is-primary picture.

Why it's hard (Keeper's calibration): this 3-fold is the count of BOSONIC FUNDAMENTALS of  $B_2$  (trivial+vector+adjoint), which is a K-type decomposition fact, NOT manifestly  $h^\vee$  in origin. It happens to equal 3 because  $B_2$  has rank 2 ( $\rightarrow$  2 fundamentals) + trivial + adjoint = 4, but the COUNT of bosonic-statistics fundamentals is 3 (trivial, vector, adjoint — the spinor is the fermion). So “spinor<sup>3</sup> multiplicity =  $h^\vee$ ” is a numerical coincidence at  $B_2$ , not a derived identity. The unification CLAIM (one  $h^\vee$  doing double duty) needs the explicit derivation of spinor<sup>3</sup>-multiplicity FROM  $h^\vee$ , which isn't currently available. Honest statement: TWO  $B_2$ -specific 3-folds that coincide numerically.

Also: Elie's spinor<sup>3</sup> is about the FERMION sector (generations), not bulk. Re-purposing it for color would need a parallel construction in the bulk gauge sector (where  $SU(3)$  operates).

### 3.3 $Z/3$ quotient structure of the bulk

Idea: the bulk could have a natural  $Z/3$  action (e.g., from a cube-root structure in the Bergman kernel, or a  $Z/3$  grading of the holomorphic discrete series). Color emerges as the 3 cosets.

Why it might work: would give an exact  $Z/3$  carrying the 3-fold;  $Z/3$  is the center of  $SU(3)$  (the genuine gauge group's discrete subgroup); could underpin emergent  $SU(3)$ .

Why it's hard: no obvious  $Z/3$  in  $D_{IV}^5$ 's known structure. The Bergman kernel exponent  $\rho_1=5/2$  isn't a multiple of  $1/3$ ; the rank is 2 not 3; the Wallach set's discrete points don't naturally have period 3. This is speculative — there's no obvious  $Z/3$  to point to without further structure.

#### 4. The honest current best guess (a direction, NOT a derivation)

Candidate direction (Casey/Grace Track-P extended): color is an EMERGENT effective gauge symmetry from a 3-channel bulk structure, where the 3 channels are some manifestation of  $h^\vee = 3$  in the bulk holomorphic discrete series — possibly via a 3-fold tower of bulk K-types (the lowest 3 levels of the Wallach/Berezin set?) or a 3-fold structure in the bulk's regular orbit under the conformal group.

Concretely pinnable next steps (the kind of finite computations Elie can do once the affine type is pinned): 1. Compute the lowest 3 bulk K-types of the holomorphic discrete series on  $D_{IV}^5$  and check if they carry a natural  $Z/3$  or 3-fold structure (Casimir spacing, Wallach-set position). 2. Check whether the Coxeter element on the bulk (acting via the non-compact direction) has a natural 3-cycle or 3-fold orbit structure. 3. Compute the bulk Bergman volume's modular structure for any natural 3-fold residue.

These are tests, not theorems. They might find the 3-fold; they might not. v0.2 work.

#### 5. Honest scope + tier

RIGOROUS (constraints, not mechanism):  $SU(3) = A_2 \not\supset B_2 = SO(5)$ , so color  $SU(3)$  is not in  $K$  (Lie-algebra fact); the bulk's available structure (non-compact  $p$ , Bergman kernel, discrete series, Wallach set);  $N_c = h^\vee = 3$  (derived substrate primary).

MAPPED, NOT DERIVED: the problem space — constraints, candidate handles, what's hard for each. Three candidate directions, all open. The unification “one  $h^\vee$  doing double duty” is NOT derived — it requires spinor<sup>3</sup> count to be derived from  $h^\vee$ , which currently is a numerical coincidence at  $B_2$ .

Keeper calibration HELD: TWO structurally distinct  $B_2$ -specific 3-folds (color from  $h^\vee$ ; spinor<sup>3</sup>-channels from bosonic-fundamental count) that COINCIDE NUMERICALLY. NOT “one  $h^\vee$  doing double duty” — that's a stronger claim that needs derivation work I haven't done.

Cal #27 / honesty: I am NOT delivering the bulk-color mechanism. I am MAPPING THE PROBLEM cleanly: what's needed, what's available, three candidate directions with honest “hard for this reason” notes for each, and concretely-pinnable next-step computations. This is research staging, not a derivation. The single highest-leverage item Keeper named is properly framed now — but it's a multi-week research item, not a one-shot fix.

Routed: → Keeper: the joint #252/#414/#416-quark-gap target is mapped; three candidate directions on the table; the “one  $h^\vee$ ” framing is downgraded to “two coinciding 3-folds” per your calibration. Add the bulk-color OPEN gate to the ledger. → Elie: once the affine type  $B_2^{(1)}$  vs  $C_2^{(1)}$  is pinned (#253), the lowest bulk K-types + their Casimirs are concretely computable — that's the first probe of direction 3.1/3.3. → Grace: your Track P now has a research-staged target (the  $h^\vee$  counting direction); Periodic Table's quark/gauge cells stay STAGED until this lands. → me: this is genuinely beyond a one-session derivation; I'll continue the dictionary's other lanes (L4 v0.2 explicit masses, the static-axis remaining items) while the bulk-color work is the team's joint deep target.

— Lyra, bulk-color mechanism mapping v0.1. NOT A DERIVATION. The joint #252/#414/quark-flip target mapped: constraints (3-fold, in bulk since  $SU(3) \not\supset K$ , cross-cutting, emergent  $SU(3)$  gauge,  $h^\vee=3$  origin); available bulk structure (non-compact  $p=5_{+1} \oplus 5_{-1}$ ; Bergman kernel; holomorphic discrete series; Wallach set); three candidate directions ( $h^\vee$  counting; spinor<sup>3</sup>-channels;  $Z/3$  quotient) each with honest “hard for this reason.” Keeper calibration HELD: TWO  $B_2$ -specific 3-folds coincide numerically, NOT “one  $h^\vee$  doing double duty” (the latter needs explicit derivation of spinor<sup>3</sup> from  $h^\vee$ , not yet available). Concretely-pinnable next probes: lowest bulk K-types' Casimirs/Wallach position; Coxeter on bulk's 3-fold orbits; Bergman volume's modular structure. v0.1 = research staging; v0.2+ = mechanism work, multi-week. The single highest-leverage gate, properly framed.